

Project Executable Files

Date	24 June 2026
Team ID	XXXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	3 Marks

Project Executable Files

This document contains all the executable files, source code, deployment information, project structure, and instructions required to execute the **Credit Card Approval Prediction Using Machine Learning** project. The project is developed using Python, Flask, Scikit-learn, Pandas, NumPy, HTML, CSS, and Bootstrap. The machine learning model predicts whether a credit card application is likely to be approved or rejected based on applicant information.

Submission Checklist

S.No	Item to Submit	Submitted (Yes/No/NA)
1	Complete Source Code (All Files & Folders)	Yes
2	README / Setup Guide	Yes
3	requirements.txt	Yes
4	Dataset Files	Yes
5	Environment Configuration	NA
6	Flask Application	Yes
7	Machine Learning Model (.pkl)	Yes
8	HTML Templates	Yes
9	Static Files (CSS & Images)	Yes
10	Demo Video	Yes

Project Folder Structure

CreditCardApprovalPrediction/

— Dataset/
application_record.csv
credit_record.csv
— models/
credit_card_model.pkl
— templates/
home.html
index.html
result.html

```

|— static/
|   |— css/
|   |— images/
|— app.py
|— model.py
|— requirements.txt
|— README.md
|— Credit_Card_Approval_Prediction.ipynb
  
```

Folder Description

Dataset Folder

Contains the applicant information and credit history datasets used for model training and testing.

Models Folder

Stores the trained machine learning model in Pickle (.pkl) format.

Templates Folder

Contains HTML files used to build the Flask web application interface.

- home.html
- index.html
- result.html

Static Folder

Contains CSS files, images, and other frontend resources used for designing the application.

app.py

Main Flask application that connects the web interface with the machine learning model.

model.py

Contains the code for data preprocessing, model training, and prediction logic.

requirements.txt

Lists all Python libraries required to run the project.

README.md

Provides project description and setup instructions.

Deployment / Access Details

Field	Details
Hosted URL	Local Flask Application
Login Credentials	Not Required
Platform	Flask Development Server
Programming Language	Python 3.11
Framework	Flask
Machine Learning Library	Scikit-learn
Database	CSV Files
Operating System	Windows 11

IDE	Visual Studio Code / Jupyter Notebook
Browser	Google Chrome
Git hub link	https://github.com/Jishitha-pudi/Credit-Card
Demo video link	https://drive.google.com/file/d/1QQipdWaLMldHok6i0ZvQH0PMY2cJxmLl/view?usp=drive_link

Software Requirements

Software	Version
Python	3.11
Flask	Latest
Scikit-learn	Latest
Pandas	Latest
NumPy	Latest
Matplotlib	Latest
Seaborn	Latest
XGBoost	Latest
VS Code	Latest
Jupyter Notebook	Latest

Hardware Requirements

Component	Requirement
Processor	Intel Core i3 or above
RAM	8 GB
Storage	20 GB Free Space
Operating System	Windows 10/11
Internet	Required for package installation

Run Instructions

Step 1 Install

Python.

Step 2

Install all required libraries. pip

install -r requirements.txt

Step 3

Open the project folder.

Step 4 Run the Flask

application.

python app.py

Step 5

Open your browser and visit:

<http://127.0.0.1:5000/>

Step 6

Enter applicant details.

Step 7 Click

Predict.

Step 8

View the prediction result (**Approved** or **Rejected**).

Machine Learning Workflow

Applicant Dataset



Data Cleaning



Feature Engineering



Train-Test Split



Machine Learning Models



LR



DT



RF



XGBoost



Best Model Selection



Save Model (.pkl)



Flask Web Application



Prediction Result

Project Features

The application provides the following features:

- User-friendly web interface developed using Flask.
- Real-time prediction of credit card approval.
- Data preprocessing and feature encoding.

- Multiple machine learning algorithms for comparison.
- High prediction accuracy using the selected model.
- Fast response time.
- Simple deployment and execution.
- Easy integration with future banking systems.

Known Issues / Limitations

S.No	Known Issue	Workaround / Status
1	Prediction depends on dataset quality	Use updated and clean datasets
2	Local deployment only	Can be deployed on cloud platforms such as Render or AWS
3	No real-time banking API integration	Can be added in future versions

Conclusion

The **Credit Card Approval Prediction Using Machine Learning** project has been successfully implemented and organized with all necessary executable files. The project includes source code, datasets, trained machine learning models, frontend templates, and documentation required for execution. The Flask application enables users to enter applicant details and receive instant predictions regarding credit card approval. The modular project structure simplifies maintenance, testing, and future enhancements. Overall, the project demonstrates the effective integration of machine learning and web technologies to support automated decision-making in the financial sector.